

Question

The following chemical processes exist:

1. Zirconium nitride (ZrN) is a relatively inert solid material, but it can dissolve in hot, concentrated sulfuric acid to form a 1:1 type ionic compound, releasing only a simple gaseous substance during the dissolution process.
2. Under a nitrogen atmosphere, a mixed solid of B_2O_3 and CaB_6 is co-heated, producing only two binary compounds, one of which can be used as a wear-resistant material.
3. Treating Tb_4O_7 with 0.1 mol/L dilute hydrochloric acid can produce pure TbO_2 .
4. Magnetite is added to an excess of $\text{K}_2\text{Cr}_2\text{O}_7 - \text{H}_2\text{SO}_4$ solution to dissolve it.
5. In an alkaline sodium hydroxide solution, nickel sulfide reacts quantitatively with carbon monoxide in a molar ratio of 1 : 5.

Given that the stoichiometric coefficients are balanced as integer ratios, the correct option regarding the chemical equations involved in the above five processes is:

- A. All other options are incorrect
- B. The sum of reactant coefficients for Process 1 is 8.
- C. The reactant coefficients of Process 2 are all less than 10.
- D. The sum of product coefficients in Process 3 is greater than the sum of reactant coefficients in Process 1.
- E. The absolute value of the difference between the sum of reactant coefficients and the sum of product coefficients in Process 4 is a prime number.

- F. The ratio of the sum of reactant coefficients to the sum of product coefficients in Process 5 is a decimal.

Answer

Correct Answer: **A**

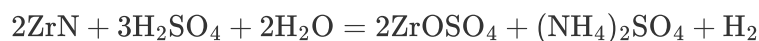
Detailed Explanation

Process 1: Zr is a congener of Ti. Similar to TiO^{2+} in concentrated sulfuric acid, ZrO^{2+} should be generated. Therefore, the product is ZrOSO_4 , which conforms to a 1:1 type ionic compound. Only elemental gas is released, which can only be hydrogen gas. Balancing the equation:

CHECKPOINT

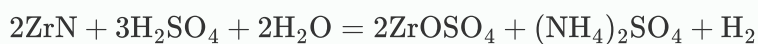
1 PTS

Zr in concentrated sulfuric acid should generate ZrO^{2+} similar to TiO^{2+}



CHECKPOINT

1 PTS



The sum of the reactant coefficients is 7, so option B is incorrect.

Process 2: Under nitrogen atmosphere, the wear-resistant material is obviously BN. According to the law of conservation of elements, the other product can only be CaO. Balancing the equation:

CHECKPOINT

1 PTS

The wear-resistant material is obviously BN



CHECKPOINT

1 PTS



The reactant coefficients include 10, so option C is incorrect.

Process 3: From the perspective of average oxidation state, Tb_4O_7 obviously undergoes a disproportionation reaction. The oxidation state of TbO_2 increases. As a rare earth element, the stable oxidation state can only decrease to Tb(III) , so the product is TbCl_3 .

From another perspective, Tb_4O_7 can be regarded as $2\text{TbO}_2 + \text{Tb}_2\text{O}_3$. TbO_2 is prepared, so the other molecular product can be regarded as Tb_2O_3 dissolved in hydrochloric acid to obtain TbCl_3 .

Balancing the equation:

CHECKPOINT

1 PTS

The other molecular product is TbCl_3



CHECKPOINT

1 PTS



The sum of the product coefficients is 7, so option D is incorrect.

Process 4: Magnetite is Fe_3O_4 . Acidic potassium dichromate has strong oxidizing properties. After dissolution, Fe(III) is generated, and the reduction product of potassium dichromate is Cr(III) . Therefore, balancing the chemical equation:

CHECKPOINT

1 PTS

Magnetite is Fe_3O_4

CHECKPOINT

1 PTS

Acidic potassium dichromate has strong oxidizing properties, and Fe(III) is generated after dissolution



CHECKPOINT

1 PTS



The sum of the reactant coefficients is 38, and the sum of the product coefficients is 42. The difference is -4, which is a composite number, so option E is incorrect.

Process 5: The reaction of nickel with carbon monoxide can be thought of as generating the famous carbonyl compound $\text{Ni}(\text{CO})_4$; it is a redox reaction. Since it is a 1:5 reaction, the remaining one molecule of CO acts as a reducing agent and is oxidized to CO_2 , which is written as CO_3^{2-} under alkaline conditions. Therefore, balancing the equation:

CHECKPOINT

1 PTS

Generating the carbonyl compound $\text{Ni}(\text{CO})_4$; it is a redox reaction

CHECKPOINT

1 PTS

The remaining one molecule of CO acts as a reducing agent



CHECKPOINT

1 PTS



The sum of the reactant coefficients is 10, and the sum of the product coefficients is 5. The quotient is an integer 2, so option F is incorrect.

Options B-F are all incorrect, so option A is correct.